

Last Name : First Name : Group :

Exercise 1 (10.5 pts): Choose the correct answer.
One or more answers may be correct for each question.
Given that 'X' is an integer variable.

1. Which instruction inputs an integer and stores it in the variable X?

- ☐ scanf ("%f", &X);
☐ printf ("%d", &X);
- ☒ scanf("%i", &X);
☒ scanf("%d", &X);

2. Which instruction displays the value of the variable X on the screen?

- ☐ printf("%f", X);
☐ scanf("%d", &X);
- ☐ printf("%d, &X");
☒ printf("%i", X);

3. When X = 5, the instruction printf ("X") will display:

- ☐ 5
☒ X
- ☐ 5.0
☐ "X"

4. Which of the following is a valid identifier?

- ☒ X1
☒ _1X
- ☐ 1X
☐ X-1

5. What is the result of the operation 5 % 2.5?

- ☐ 0
☐ 2.0
- ☐ 2
☒ Error

6. To increment X by 1:

- ☒ X=X+1
☒ X+=1
- ☒ X++
☐ X=+1

7. In an if...else instruction, when the condition is false:

- ☐ The program stops
☒ The else block is executed
- ☒ The if block is ignored
☐ Error

8. How many times will "hello" be displayed?

X = 1;
while (X != 10) {
 printf ("hello");
 X = X + 2; }

- ☐ 0
☐ 5
- ☐ 1
☒ An infinite number

9. Which loop guarantees that the code block executes at least once?

- ☐ for
☒ do-while
- ☐ while
☐ nested loop

Exercise2 (5.5pts):

A shop offers discounts to its customers, depending on the amount of their purchases (A).

Amount of purchases (DA)	Discount
A < 2000	5%
2000 ≤ A < 5000	12%
A ≥ 5000	20%

Write a C program that:

- Inputs the amount of a customer's purchases (A).
- Calculate and display the discount (D) and the price to pay (P).

// Solution – Exercise 2

float A, D, P;

// Input purchase amount

printf("Enter the purchase amount (DA): ");

scanf("%f", &A);

// Calculate discount

if (A < 2000) {

 D = A* 0.05;

} else if (A < 5000) {

 D = A*0.12;

} else {

 D = A* 0.20;

}

// Calculate and display price to pay

P = A - D;

printf("Price to pay: %.2f DA ", P);

20

